

Jiaming Yao

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EDUCATION

University of Copenhagen

Astrophysics, Master of Science

Copenhagen, Denmark

09/2020 - 06/2022

- **Thesis:** The Optical and Infrared Properties of Dusty Star-forming Galaxies in ALCS | Supervisor: Prof. Francesco Valentino, Prof. Georgios Magdis

Northeastern University

Applied Physics, Bachelor of Science | GPA: 3.02/5

Shenyang, China

09/2016 - 06/2020

- Thesis: Time Evolution of Merger Rates of Primordial Black Holes | Supervisor: Prof. Nan Li

RESEARCH EXPERIENCE

Student

DAWN, Niels Bohr Institute

08/2021 – 06/2022

Copenhagen, Denmark

- Master thesis project on dust in lensed galaxies.

Research Assistant

Institute of High Energy Physics

08/2019 – 09/2019

Beijing, China

- Searched for and recorded signals of gamma-ray bursts in light curves from the Hard X-ray Modulation Telescope.
- Reported GRB candidates to the Gamma-ray Coordinates Network.

CONFERENCE EXPERIENCE

Presenter

DAWN Summit

09/2021

Copenhagen, Denmark

- Gave a brief talk about project on distinguishing red quiescent galaxies and blue galaxies by use of UVJ diagram.

Presenter

Annual Danish Astronomy Meeting

06/2022

Frederica, Denmark

- Introduced Master's work on poster and by giving a talk.
- Actively engaged in post-presentation discussions, answered questions, and facilitated dialogue to deepen understanding and exchange idea

OBSERVATIONAL EXPERIENCE

Student

NOT Summer School

08/2021

La Palma, Spain

- Selected objects to be observed according to interest and visibility plots, then made a catalog of these objects including coordinates and magnitudes.
- Used the ALFOSC and FIES of the Nordic Optical Telescope to carry out observation.
- Retrieved images and spectra from the server, used scripts to process raw data.
- Produced clean images of nebulae and galaxy mergers. Processed and obtained clean spectra of supernovae, novae, and a white dwarf. Learned different features of these celestial objects, like hydrogen emission lines in spectra of novae and strong UV emission lines in the spectrum of the white dwarf.

RESEARCH PROJECTS

Properties of dust in lensed distant galaxies | *ALMA, modified black body*

08/2021 - 06/2022

- Searched for counterparts of ALMA-detected objects in catalogs of the ALMA Lensing Cluster Survey(ALCS).
- Used sub-mm data and modified blackbody to model spectral energy distributions of star-bursting candidates and from which deduced physical properties of galaxies like dust masses and star formation rates. Tested common bi-variate relationships such as $SFR-M$ and $L_{IR}/L_{UV}-M_{\star}$.
- Conducted visualization based on different redshift bin and discussed possible mechanisms of different groups or outliers.

OTHER EXPERIENCE

Student

06/2022

Summer School in Astrostatistics

Penn State University, US

- Studied maximum likelihood estimation, Bayesian inference, and basic concepts in machine learning.
- Learnt and practiced common statistical functions and packages in R.

TECHNICAL SKILLS

Languages: English, Chinese

Programming languages: Python (intermediate), R (beginner)

OS: Unix, DOS

Document softwares: LaTeX, Office

Certifications: Bayesian Statistics: From Concept to Data Analysis (Coursera)